

[Lecture Note] Secondary Data

GE2021 Research Technology, Spring 2025

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Learning Objectives

- 8-1 Describe the benefits of using secondary data.
- 8-2 Identify the major sources of secondary data.
- 8-3 Explain the drawbacks of using secondary data.

SHORT VIDEO INTRODUCTION

<https://www.youtube.com/watch?v=CPQ95B95bWE&t=101s>

Benefits of Using Secondary Data

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- Secondary data: data or raw information collected by other researchers.
 - Primary data: data collected for a specific purpose by the researcher.

Availability of Data

- Researchers may be interested in data or information that has become impossible to collect.
- Accessing secondary data allows the researcher to have information that would have never been directly available to them.

Availability of Design and Sampling Information

- When data is available from a previous study, the researcher also has all the details on how the study was conducted, designed, and what sampling type was selected.
- This frees the researcher from having to make such decisions, but it is still important that they become familiar with the design, sampling, and data collection process, because the design is related to the data analysis.

Opportunities for Replication

- Replication of a study can happen in two ways:
 - Collect primary data that are exactly like the original study, but in a different population or location to see whether the findings hold true
 - Use secondary data by collecting additional information that may be missing from the original study.
 - Draw on the original study to learn additional information from the population
 - Data can be collected from the same participants or from the same type of population.

Protection of Participants

- Secondary data can be useful with sensitive populations of interest.
- Interviewing a participant and asking them to recall difficult events can cause emotional and psychological harm, so using secondary data such as police reports, medical examinations, and so on, can protect the participants.

Time Effectiveness

- Collecting primary data requires a great deal of time, especially for larger studies.
- Analyzing already-collected data is much easier, and the researcher is able to move directly to cleaning, coding, and analyzing the data.
 - Often secondary data is already clean, without missing values or other issues researchers must resolve when working with primary data. ## Cost Effectiveness
- Primary data collection requires a large budget that may not always be available.

Large Data Sets

- Secondary data allows the researcher to use a large amount of data that would not have been otherwise possible.
- Larger data sets allow the researcher to analyze huge amounts of data, which make it easier to generalize their findings to the general population.

- Data scraping: the process of extracting large amounts of information from social media, business sites, or any other websites of interest into a readable spreadsheet.

Major Sources of Secondary Data

- Although there is an abundant amount of secondary data available, it is important to carefully examine the secondary sources by looking at their methodologies.
- The researcher also needs information on data availability, such as the possibility of using the data for their own analysis and whether there are applicable terms and conditions.

Government Statistics

- All governments collect various forms of information about their citizens.
- These large databases are available to researchers in raw form and include basic information.
- Government statistics are some of the most comprehensive and oldest data available for most populations.
- Data sets are available from government agencies such as the Census Bureau, the Centers for Disease Control and Prevention, the National Archives, the Department of Education, and so on.

Research University Data

- Research universities often create institutes dedicated to data collection in a specific field that make valuable longitudinal data available to other researchers.
- One of the largest resources of data sets available to students is the Inter-university Consortium for Political and Social Research (ICPSR)
 - The ICPSR is hosted by the University of Michigan and is a collaboration of 776 institutions, mostly academic, but also government agencies and others.
- Another such resource is the General Social Survey (GSS), which has been collecting data since 1972 and has sister branches internationally.
- Another option is the data collected by a specific researcher at a research university.
 - Many researchers want other researchers to analyze their data and are eager to provide all the information needed if they are asked, but this remains their prerogative.

Institutional Data

- Any large organization, such as schools, hospitals, large businesses, and so on, collects information about their clients and employees.
- Their data are often in raw form and not collected for research purposes, but they are still a potential source of data for researchers.

Online Sources

- Social media hubs have demographic data and data pertaining to specific issues.
- Online reviews of products, procedures, events, and so on, can also provide information.

Disadvantages of Secondary Data

Uncertainty of Constructs

- New studies and findings continually improve previous construct measurements, so using secondary data can limit a study to the results of an older or possibly outdated data collection method.
- The original study may have had a different focus, so some constructs may have been more important to the original researchers than it is to the new researcher.
- Researchers must consider whether the original construct fits the focus of their current study; uncertainty about the measurement of the original construct can cause flaws in the new study's results.

Ambiguity of Measurement Error

- When a researcher is highly involved in the process of data collection, they are aware of every aspect of the process, which is crucial to data analysis.
- In using secondary data, researcher often take the information at face validity because there is a lack of good sense of measurement error.

Passage of Time

- There is a possibility that secondary data is outdated.
- Depending on the study's focus, some information becomes old quite quickly, whereas other information may remain relevant.
- However, the passage of time can be beneficial if the researcher is interested in longitudinal studies and wants to see how a concept has changed over time.

Class Activities:

- The American Community Survey (ACS) is a great source of secondary data on populations and communities. This activity will provide you with an opportunity to review data from your own community. Follow the steps below:

- (a) Visit the Census Bureau's American Data website at <https://data.census.gov/cedsci/>

- (b) In the search box, type in your community zip code and click search
 - (c) Find the following facts for your respective community by navigating the toolbar at the top of the search page (All, Tables, Maps, and Pages):
 - * Total population
 - * Median age
 - * Educational attainment (high school graduate or higher)
 - * Total housing units
 - * Median household income
 - * Individuals below poverty line
- Share this information with other students in your class and compare and contrast your community statistics. Discuss the potential limitations of this data as a class.

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- The Centers for Disease Control (CDC) is a great resource for data on key health indicators. This activity will provide you with an opportunity to learn more about these key health indicators.
 - (a) Visit the CDC website at <https://www.cdc.gov/DataStatistics/>
 - (b) Click on “Diseases and Conditions” in the toolbar at the top of the page
 - (c) Click on “Data and Statistics” in the drop down menu that appears
 - (d) Pick one disease topic interest to you from the list of Diseases and Conditions
 - (e) Based on the data that is presented to you, respond to the following questions:
 - * Number of people affected by this problem (morbidity statistics)
 - * Populations disproportionately affected by this problem (disparities)

Please note, each disease topic page is organized differently. This exercise is intended to give you an opportunity to browse the information included on this secondary data resource.

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- There are various online sources of data, including data warehouses such as the ICPSR website. This activity will provide you with experience in accessing and downloading data.
 - Click on <http://www.icpsr.umich.edu/icpsrweb/ICPSR/>
 - Click on “Find Data” in the toolbar
 - Pick a topic area of interest to you and type it into the box (e.g., suicide)
 - Click on one of the data sets the website identifies for you
 - Read the study “Summary”
 - Under “Table of Contents” on the left hand side of the page, click on “Methodology”
 - Read and briefly summarize the study methods
 - Share what you have learned about this study goals and methodology with students in your class

House Keeping

Announcement

Assignment

Assignment1: Assignment Title

- What to do

- A
- B

- Format and Requirement

- PDF format, < ?? pages
- file name should be include your student id and name
 - * stuID_name_title.pdf (e.g. 1111111_ChungilChae_SelfIntroduction.pdf)
- APA, Double-Space, No title page, 12 points, Reference section if necessary
- Chinese name in English, English name, Student ID, Class name and section (e.g. GE2021, W09; MGS3001, W01)

- due date

- by Date(DAY) 11:59PM
- NO LATE SUBMISSION ALLOWED!!!!

Reference